Path-planning algorithms

Robotics Internship

Karunya Institute of Technology and Sciences

July 25th 2025

Contents



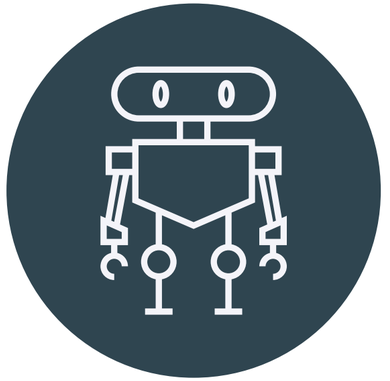
Documentation



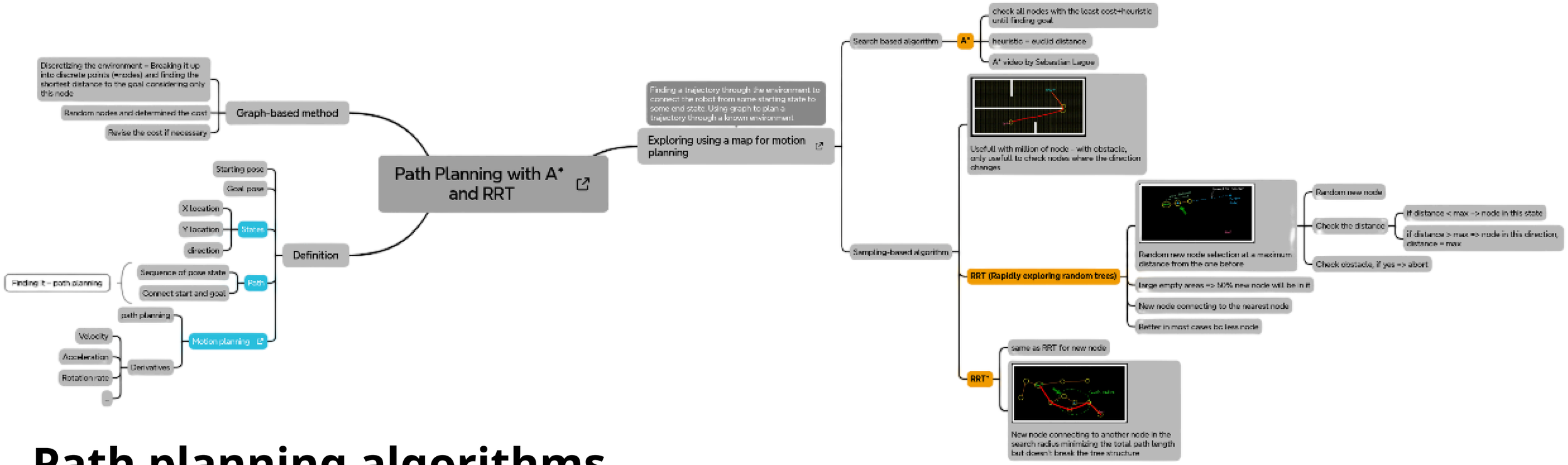
Matlab simulation



Matlab & C++



Robot Deployment



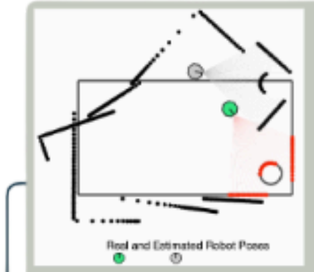
Path planning algorithms

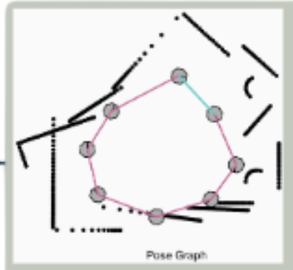
Brian Douglas - Matlab video

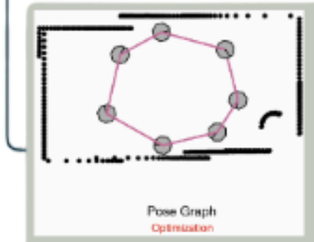
Daniel K., Nash A., Koenig S., Felner A. (2010) *Theta*: Any-Angle Path Planning on Grids*

Cheng Yi, Xiao Hongtu (2019) *Global Dynamic Path Planning Based on Fusion of Improved A* Algorithm and Morphin Algorithm*

Documentation

- SLAM Algorithm Using Pose Graph Optimization
  - Localisation and estimated pose
    - 

      Error in the odometry measurement
    - 

      Pose graph build
      - Finding pose closure
        - "Extracting Statistical Signatures of Geometry and Structure in 2D Occupancy Grid Maps for Global Localization" by Su-Yong An and Jaeyoung Kim (2022)
        - "A fast, complete, point cloud based loop closure for LiDAR odometry and mapping" by Jiarong Lin and Fu Zhang (2019)
    - 

      Optimization
  - Mapping
    - Analyzing the scans
    - Build a binary map

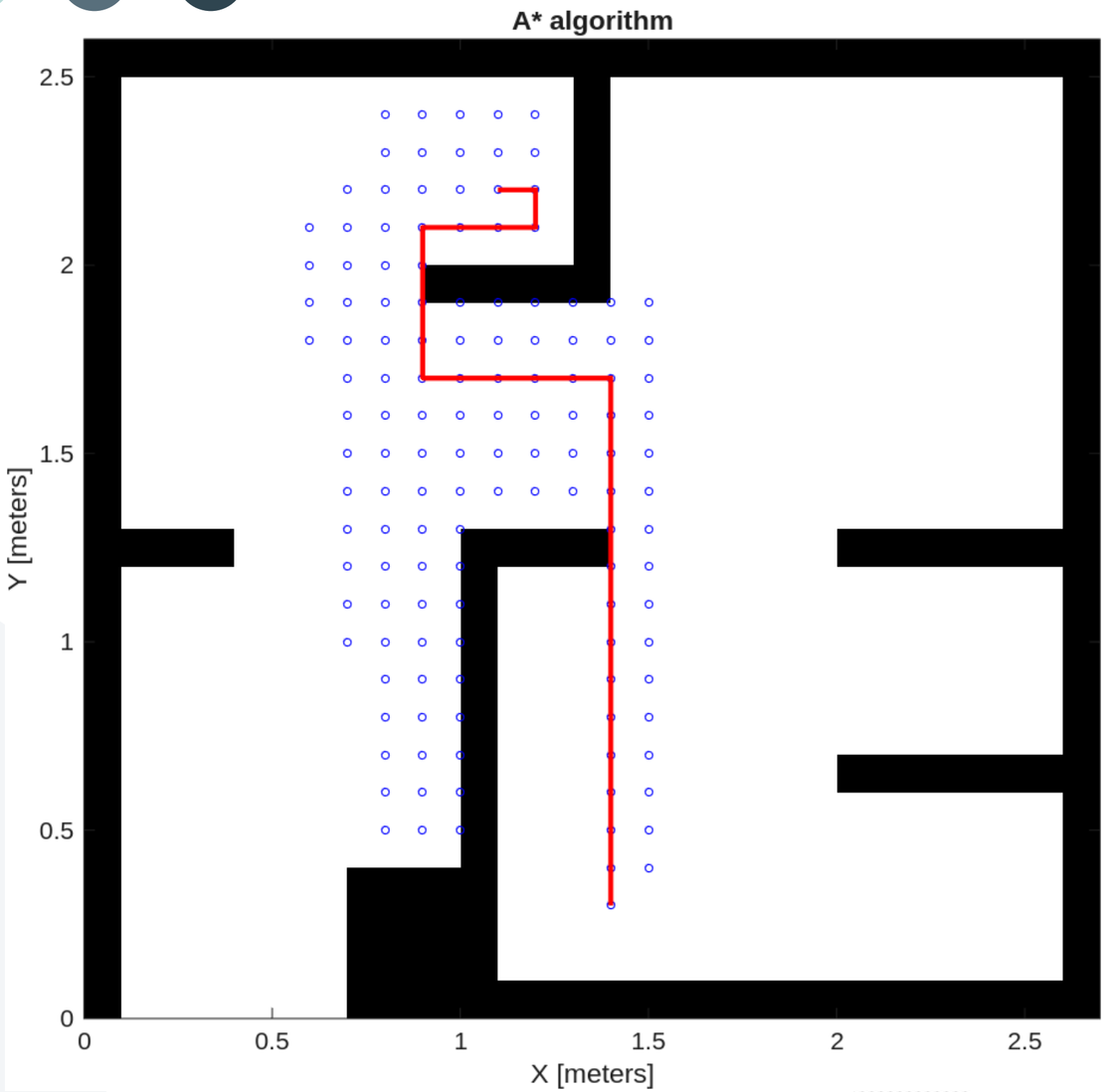
xmind

Matlab simulation

A*

Easy to implement

Not the optimal path
Too costly to execute
No control on the parameters

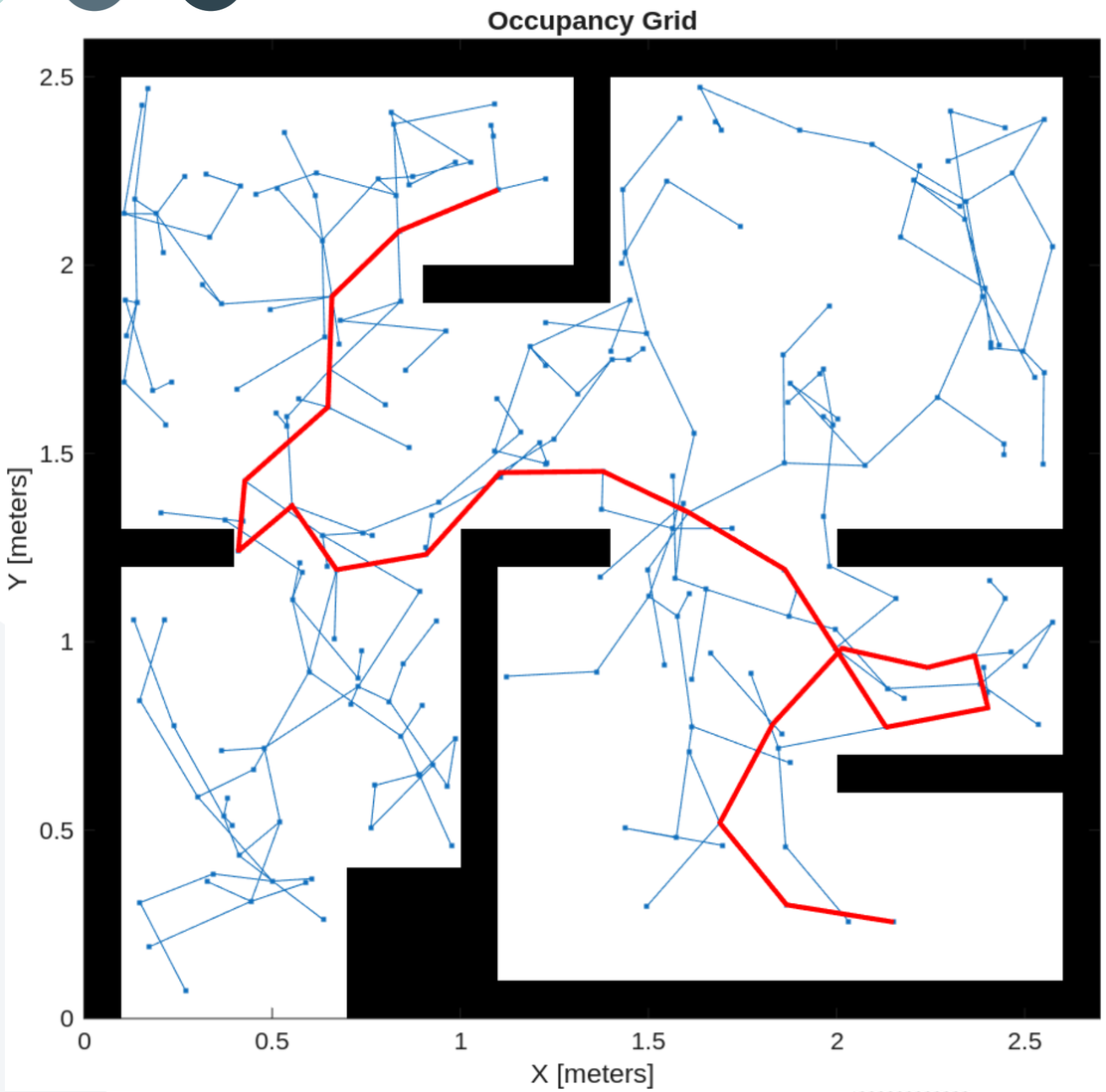


Matlab simulation

RRT

Easy to implement

Not the optimal path
Too costly to execute
No control on the parameters

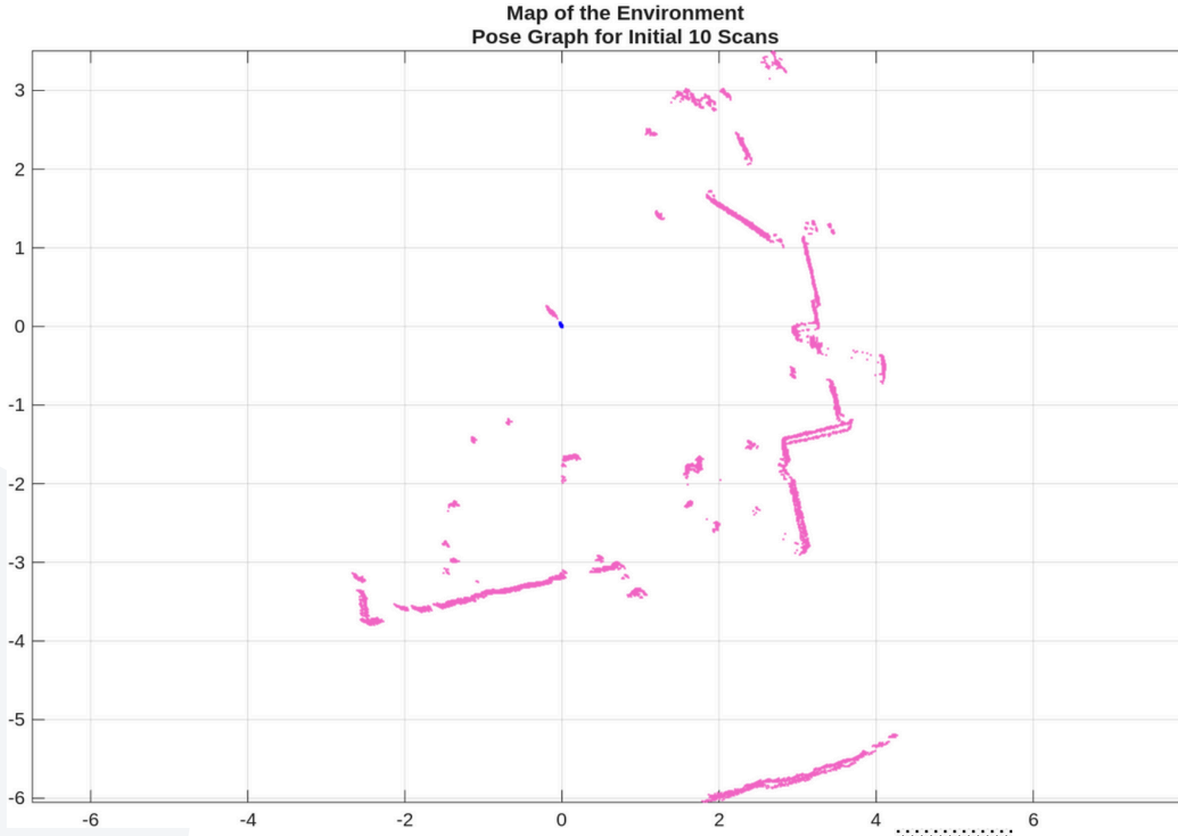


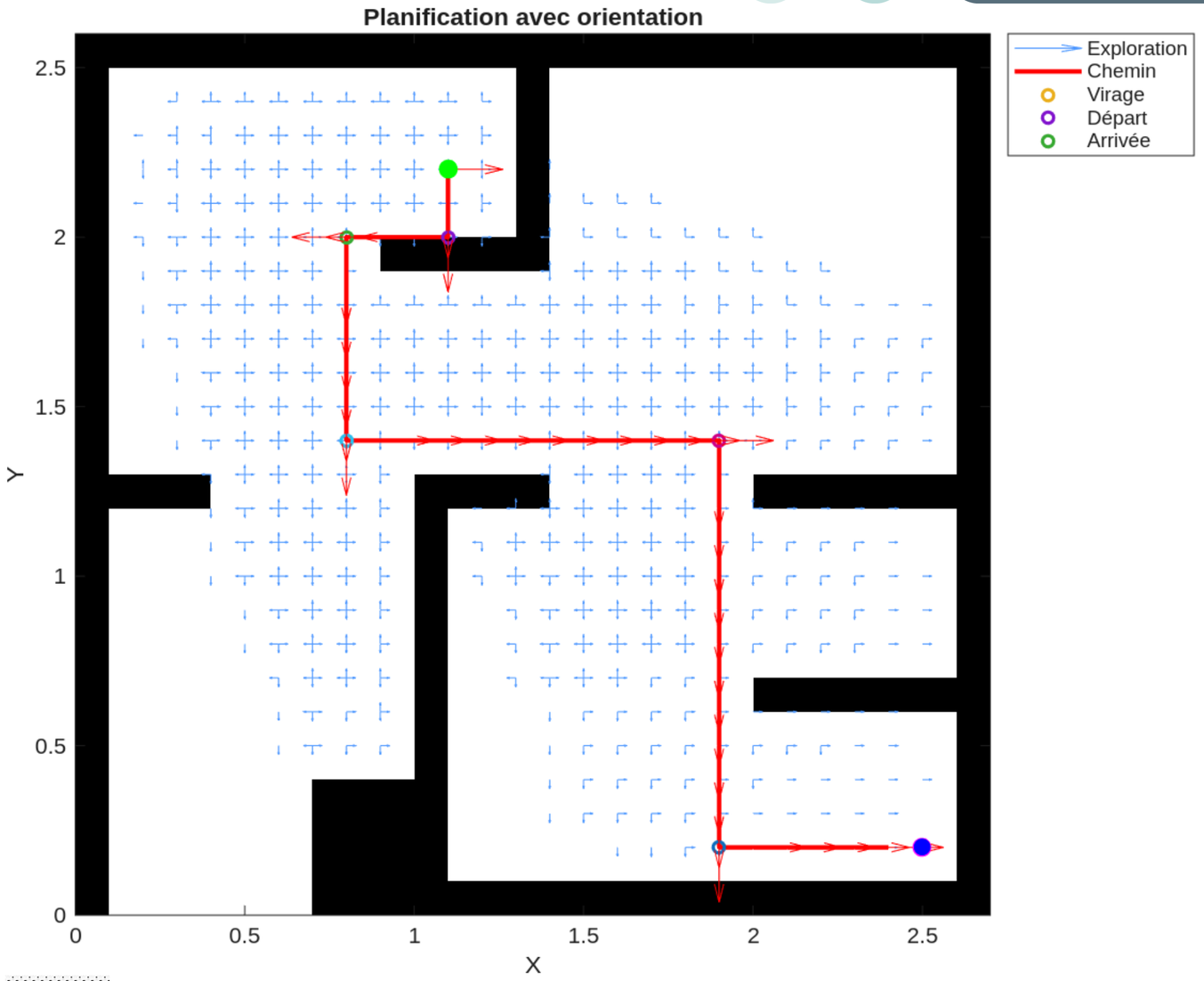
Matlab simulation

SLAM

Easy to implement

Not the optimal path
Too costly to execute
No control on the parameters





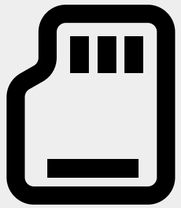
Matlab & C++

A*

More efficient execution
Optimal garantee

Matlab & C++

A* worst-cased complexity



Space Complexity

Worst-case: **O(N)**

Stores all:

- Open nodes
- Closed nodes
- Cost and parent info



Time Complexity

Worst case : **O(N*log(N))**

N: number of explored states
Comes from using a priority queue (log(N))

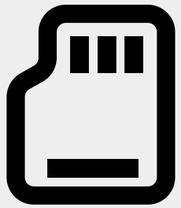
High-dimensional, **Continuous** spaces

⇓

Exponential growth in the number of states

Matlab & C++

RRT average complexity



Space Complexity

O(n)
Tree with n nodes



Time Complexity

O(n log n)
with nearest-neighbor
search structure like k-d tree

Probabilistically complete

RRT* :

- Improve the solution
- Higher complexity than RRT

Matlab & C++

Improved algorithms

RRT*

Theta*

Fusion of Improved A* Algorithm and Morphin Algorithm



Matlab & C++

SLAM

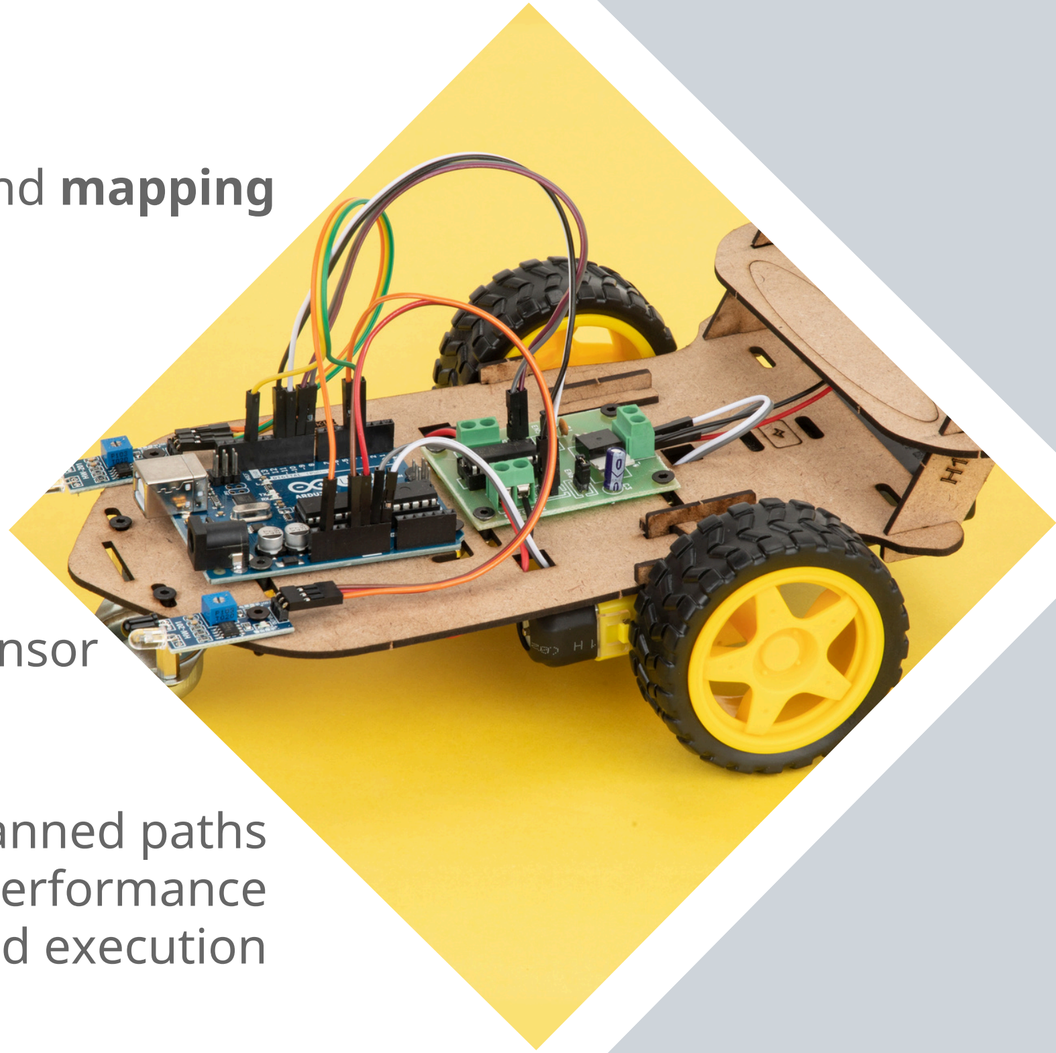
Dataset : Intel Research Lab
1361 scans with 180 points each

Pose graph construction too costly

Improving loop closure detection
with hystogram

Robot deployment

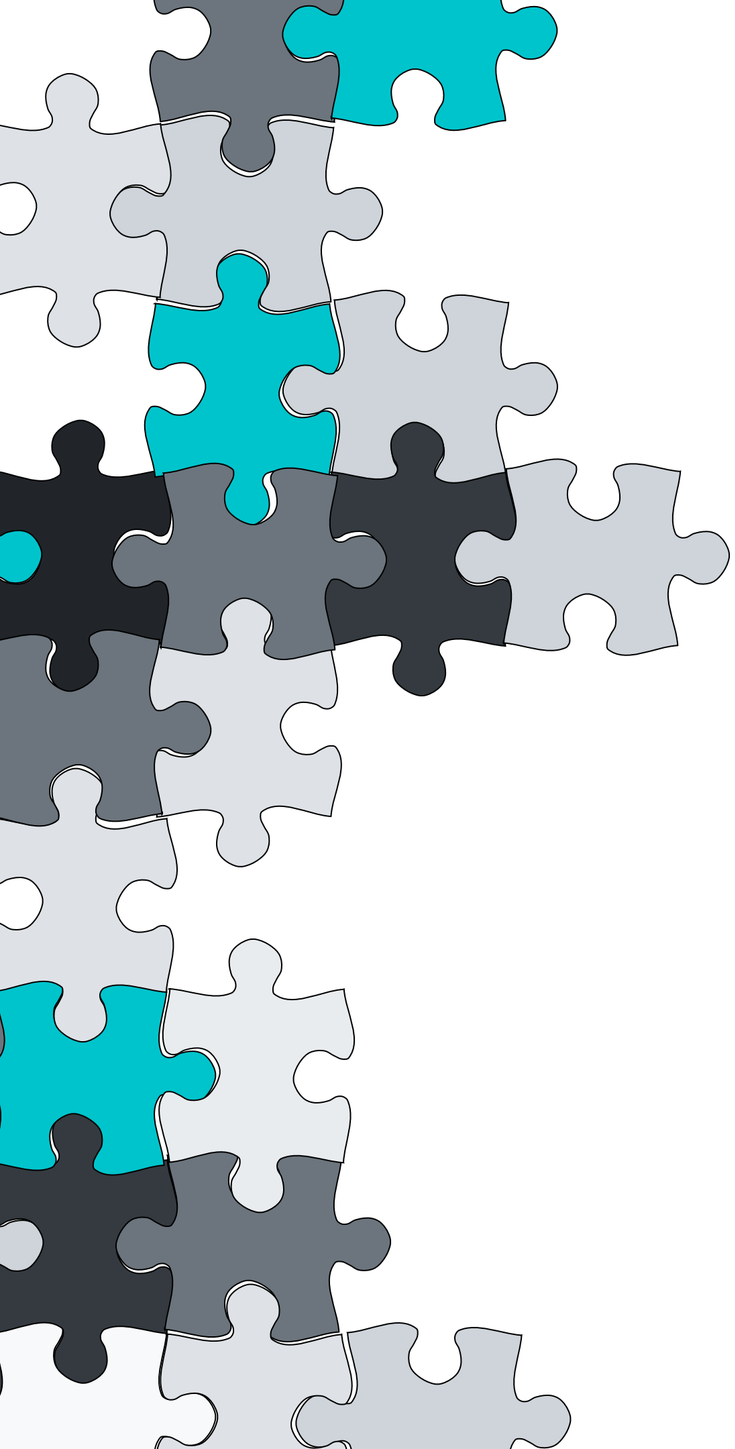
Integrate the algorithm into the robot's control system
Use live sensor data (e.g., LiDAR / odometry) for **localization** and **mapping**
Send **planned paths as motion commands** to the robot
Record and evaluate the robot's behavior during execution



Testing Environment

- **Indoor** test map with **known obstacles**
- Real-time feedback via **onboard sensors**
- Testing with known and new obstacles detect by onbord sensor

Validate the correctness and feasibility of planned paths
Observe differences between simulation and real-world performance
Identify potential improvements in robustness and execution



Thank you for your attention